

Short-Term Stock Portfolio Optimization for Retail Investors

Team: BiznesDengi

Data Mining Project, 2026





Business Objective

Goal: Support short-term investment decisions for retail investors with a 6-month horizon.

Key Objectives:

- **Return Target:** Achieve at least 11.2% return over 6 months.
 - **Rationale:** Outperform Russia's 5.6% inflation (2025) by 2x to ensure real purchasing power growth.
- **Risk Constraint:** Maximum acceptable loss of 20% of the initial deposit.
 - **Rationale:** Moderate risk tolerance, benchmarked against a 23.38% drawdown in a comparable strategy.

Success Criteria: A strategy is successful if it meets both the return and risk targets.



Situation Assessment

Resources & Constraints:

- **Team:** 5 members (Project Manager, Business Analyst, Data Scientist, ML Engineer).
- **Data:** 249 most liquid Russian stocks from MOEX ISS API (16 years of history).
- **Timeline:** Project duration of 2 months.

Key Risks & Mitigation:

- **Data Quality:** Addressed via thorough initial verification and preprocessing.
- **Model Overfitting:** Addressed via walk-forward validation and cross-validation.
- **Market Shifts (Crises):** Addressed via risk management rules (e.g., stop-loss) and explicit communication of model limitations.



Cost-Benefit Analysis

Total Estimated Project & Promotion Cost: \$15,617

- Labor (5 members x 2 months): \$10,000
- Infrastructure (Laptops): \$417
- Data Acquisition (MOEX): \$500
- Operational Costs: \$200
- Marketing (3 months): \$4,500

Break-Even Point: ~560 Clients

- **Model:** 5% commission on positive client returns.
- **Calculation:**
 - Avg. Client Investment: \$5,000
 - Target Return (11.2%): \$560 profit/client
 - Business Revenue (5%): \$28/client
 - $15,617 / 28 \approx 558$ clients



Data Mining Goals & Project Plan

Data Mining Goal: Predict the 6-month forward total return (price + dividends) for each stock.

- **Task Type:** Regression.
- **Primary Metric:** Mean Absolute Error (MAE).
- **Secondary Requirement:** Model interpretability for business trust.

Project Iterations:

- We planned **two modeling iterations** to allow for learning and improvement.
- **Iteration 1:** Baseline single-horizon model.
- **Iteration 2:** Improved multi-horizon ensemble model.



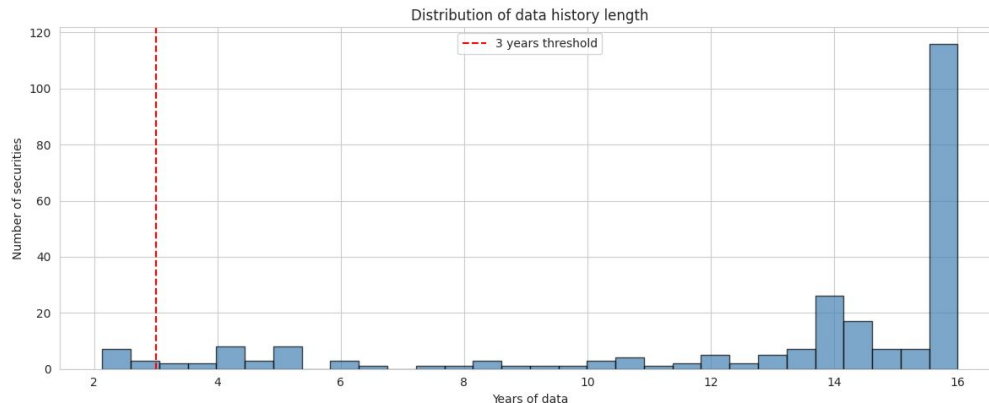
Data Understanding - Coverage & Selection

Data Selection: Parsed MOEX data for 249 ordinary shares, Feb 2010 – Feb 2026.

- Last year held out for final testing.
- Includes diverse market regimes (2014 sanctions, 2020 crash, 2022 volatility).

Key Finding: Data History Depth

- **Analysis:** A wide distribution of history length exists across securities.
- **Action:** A **3-year minimum history threshold** was defined.
 - **Rationale:** Minimum lookback needed to calculate 36-month momentum features without instability.

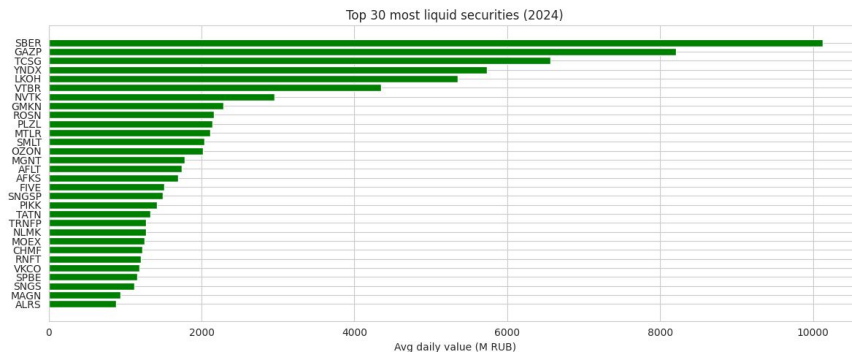




Data Understanding - Liquidity & Survivorship

Key Finding: Concentrated Liquidity

- A small group of premier stocks dominates trading volume.
- **Action:** A liquidity filter of **1M RUB minimum average daily value** was defined, retaining 215 securities for a feasible investment universe.



Key Finding: Delisted Securities

- 16 of 247 securities were delisted by Jan 2025.
- **Risk:** Survivorship bias. A model trained only on survivors will not learn from failures.
- **Action:** All securities (including delisted) were retained in the training data.



Data Quality & Preparation Overview

Data Quality Verification (Raw Data):

- **0.0% Missing Values** in critical price/volume columns.
- **Logical Integrity:** No violations of $LOW \leq HIGH$, etc.
- **Conclusion:** No complex imputation or cleaning was required.

Data Preparation Steps:

1. **Outlier Mitigation:** Clipped extreme feature values (0.1st & 99.9th percentiles) and implausible target returns.
2. **Feature Engineering:** Constructed 4 groups of features.
3. **Integration:** Merged all data sources into a single table.
4. **Final Formatting:** Created a tidy table for modeling.



Feature Engineering (1/2)

Objective: Create a rich set of predictors for the model.

Category	Features	Rationale
Technical	Momentum, Volatility, Max Drawdown	Capture price trends, risk, and downside risk at 6, 12, 24, and 36-month horizons.
Financial	Avg. Trading Value, Relative Strength, Excess Return, Beta	Measure liquidity, market-relative performance, and systematic risk.
Financial	12m Yield, 3y Avg. Yield, Consistency	Quantify dividend income, a key component of total return for 163/247 stocks.
Static / Macro	Log Issue Size, CBR Key Rate	Company size proxy and macroeconomic context.



Feature Engineering (2/2)

Target Variable: Forward Total Return

The formula captures total shareholder return:

Target Return = (Future Price - Current Price + Dividends Paid) / Current Price

Horizons: 14-day, 42-day, and 182-day (primary 6-month target).

Final Dataset Snapshot:

- **Rows:** 274,534 observations
- **Columns:** 30 (19 features, 3 targets, identifiers)
- **Securities:** 143 (reduced from 247 due to feature requirements, not data quality).



Modeling Strategy

Chosen Algorithm: XGBoost

- **Why?** Captures non-linear relationships and feature interactions without explicit specification. Robust to feature scaling and includes regularization to prevent overfitting.

Modeling Evolution:

- **Iteration 1 (Baseline):** A single XGBoost model predicting the 180-day return.
- **Iteration 2 (Final Model):** A Multi-Horizon Ensemble.
 - Trains **three separate XGBoost models** for 14-day, 42-day, and 182-day horizons.
 - **The Math: Ensemble Prediction**
 - Final Prediction = $\sum (W_h * P_h)$
 - Prediction P_h made by the individual XGBoost model for a specific horizon h
 - Weight $W_h = (1 / MAE_h) / \sum (1 / MAE_k)$
 - **Logic:** The weight for each horizon is inversely proportional to its historical error. More accurate models get higher weights.



Test Design: Walk-Forward Validation

Goal: Simulate real-world deployment and prevent look-ahead bias.

Data Partitioning (Time-Sorted):

1. **Scaler Fit:** 2015-2018 (Fit standardizer)
2. **Train/Validate:** 2018-2024 (Expanding window)
3. **Hold-Out Test:** 2024-2025 (Final evaluation)

Walk-Forward Process:

- The model is retrained every 14 days using a 3-year rolling lookback window.
- Predictions are always made on future, unseen data.
- Errors are tracked dynamically to update ensemble weights.



Model Assessment

Primary Metric: Mean Absolute Error (MAE)

Model	Horizon	Final MAE	Ensemble Weight
Iteration 2	14 days	0.0670	60.7%
	42 days	0.2717	29.2%
	182 days	0.5734	10.1%
Iteration 1	180 days	0.8380	—

Conclusion: The multi-horizon ensemble significantly outperforms the baseline. Short-term models are far more accurate and dominate the ensemble, with a small long-term adjustment from the 182-day model.



Model Assessment - Feature Importance

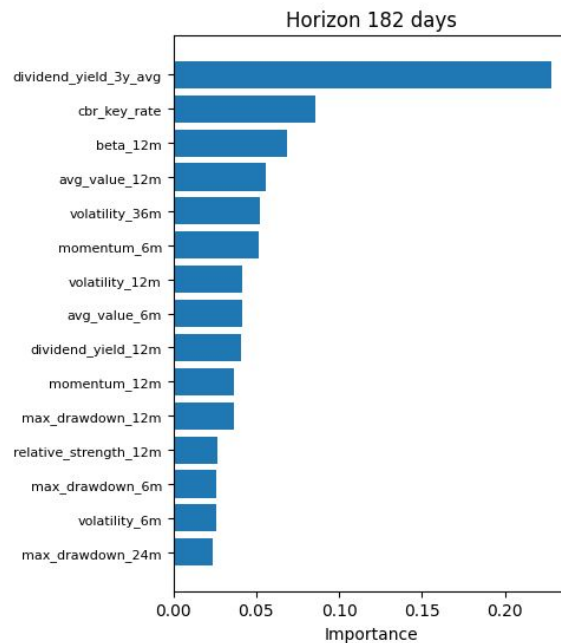
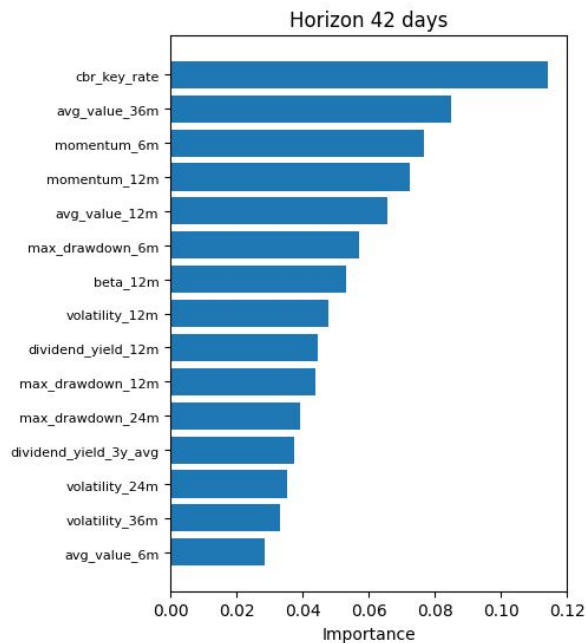
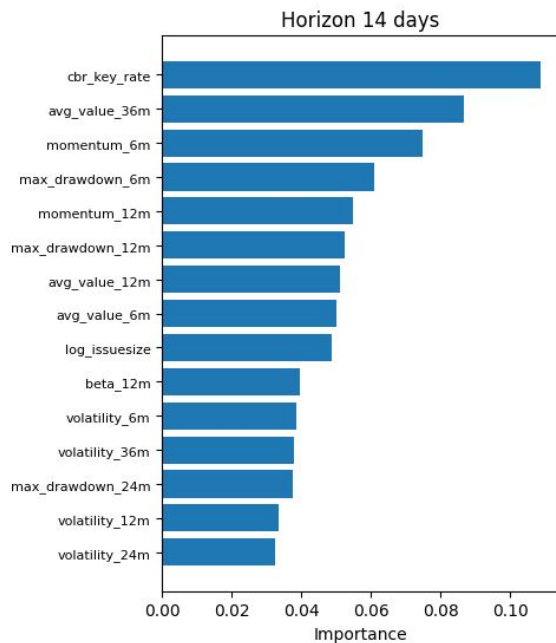
Key Drivers of Stock Returns (from Iteration 2):

1. **Macroeconomic Dominance:** The CBR Key Rate is the single most important predictor across all horizons. Monetary policy heavily influences equity returns.
2. **Liquidity Signal:** Long-term avg. trading value (36m) is the second most important feature, capturing fundamental company quality and accessibility.
3. **Short-Term Price Dynamics:** momentum_6m and max_drawdown_6m are the strongest price-derived predictors.
4. **Redundant Indicators:** relative_strength_12m and beta_12m had near-zero importance, suggesting the model captures these dynamics internally.



Model Assessment - Feature Importance

Feature Importance by Horizon





Evaluation - Portfolio Construction & Testing

From Predictions to Portfolio:

1. **Filter:** Select all stocks with a positive ensemble prediction.
2. **Diversify:** Apply HDBSCAN clustering on the stocks to group them into segments.
3. **Weight:** Assign portfolio weights proportional to the prediction score, picking one stock per cluster to ensure diversification.

Forward-Looking Test:

- Portfolios were constructed on 5 specific dates (Aug 2024 – Jun 2025) and held for a 6-month period to measure realized returns.



Evaluation - Results vs. Business Success Criteria

Portfolio Date	Number of Stocks	Realized 182-Day Return
2024-08-05	8	+0.08%
2024-09-03	8	-0.82%
2025-04-03	18	+15.59%
2025-05-05	9	+25.16%
2025-06-06	9	+10.01%

Assessment:

- **Risk Criterion Met:** Maximum loss was -0.82%, well within the 20% limit.
- **Return Criterion Partially Met:** The target return at least 11.2% return was achieved in the majority of test cases, but not consistently.



Portfolio Examples

Table X: Portfolio Composition (2025-05-05)

Ticker	Prediction Score	Weight	Cluster	Target Return
TRNFP	0.3325	22.06%	-1	14.79%
TASBP	0.2427	16.10%	0	30.54%
GMKN	0.2037	13.52%	-1	26.66%
NNSBP	0.1587	10.53%	0	23.09%
NNSB	0.1375	9.13%	0	28.11%
STSB	0.1333	8.85%	-1	—
MISBP	0.1045	6.93%	1	—
VRSBP	0.0996	6.61%	1	—
RTSBP	0.0947	6.28%	1	—

Portfolio Return: 25.16%

Table VIII: Portfolio Composition (2024-09-03)

Ticker	Prediction Score	Weight	Cluster	Target Return
TRNFP	0.2333	29.79%	-1	30.65%
NNSBP	0.1276	16.29%	0	-12.66%
IGSTP	0.0969	12.38%	-1	-28.24%
MISBP	0.0927	11.84%	0	1.33%
KCHE	0.0922	11.77%	0	-17.16%
RTSB	0.0880	11.24%	-1	—
USBN	0.0302	3.86%	1	—
UTAR	0.0222	2.84%	1	—

Portfolio Return: -0.82%

Two examples from the forward-looking test. The best case demonstrates the return potential when predictions are accurate. The worst case demonstrates how risk controls limit losses when predictions are not. A dash (—) means the stock was delisted or untraceable; its outcome is unknowable. Because no portfolio relied on any single stock, these missing outcomes did not derail overall performance.



Process Review

Strengths:

- **Traceability:** Every decision (features, filters) links back to a finding in Data Understanding.
- **Iterative Improvement:** The model significantly improved from Iteration 1 to 2.
- **Honest Reporting:** Both positive (+25.16%) and negative (-0.82%) results are shown.

Weaknesses / Areas for Improvement:

- **No Simple Baseline:** XGBoost performance wasn't compared to a linear model or historical mean predictor.
- **Small Evaluation Sample:** Only 5 test dates provide limited statistical confidence.
- **Unused Techniques:** Initial ideas like time series decomposition and neural networks were not explored.



Final Decision & Next Steps

Decision: The model is not approved for deployment to real clients due to inconsistent returns across evaluation dates. It is approved for continued development.

Recommendation: Conduct a Third CRISP-DM Iteration with targeted improvements:

1. **Expand Test Evaluation:** Generate portfolios at monthly intervals (12-18 points) to properly assess return consistency.
2. **Implement Baseline Models:** Use a naive mean predictor to quantify the true value added by XGBoost.
3. **Analyze Regime Dependence:** Segment performance by CBR key rate trend (rising/falling) to test if model performance is stable.
4. **Test Portfolio Rules:** Check if the complex HDBSCAN clustering adds value over a simple top-N strategy.